



OPEN ZnO/WO₃ composite for efficient photocatalytic degradation of methylene blue dye under solar light

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In this study, a ZnO/WO₃ composite structure has been formed using a hydrothermal approach for the potential photocatalytic degradation of Methylene blue (MB) dye, one of the most frequent colorants found in textile wastewater. The main goal of the study was to investigate the impacts of ZnO content on the structural, morphological, and photocatalytic properties of the ZnO/WO₃ composite structure. It was found that 5% ZnO/WO₃ composite structure exhibited the highest photocatalytic efficiency under simulated solar light irradiation, achieving approximately 93.8% degradation of MB dye within 60 min (volume = 30 mL, catalyst mass = 3 mg, MB concentration = 5 ppm). The observed superior photocatalytic performance of the catalyst can be attributed to the optimal synergy between ZnO and WO₃, which facilitated the improved light absorption and enhanced charge separation. It also highlights the potential of ZnO/WO₃ composite structures for water treatment applications, offering an effective approach for the degradation of hazardous dye pollutants under solar light irradiation.

Keywords ZnO/WO₃, Methylene blue, Photocatalysis, Heterojunction, Water treatment

The textile industry is one of the largest consumers of water, requiring an estimated 100–200 L of water to produce just 1 kilogram of textile product¹. After textile mills treat their fabrics with dyes such as MB, the resulting wastewater is often discharged into natural water bodies, negatively impacting water quality and disrupting ecosystems. Existing literature indicates that approximately 15% of non-biodegradable textile dyes are released into natural water sources annually². MB is widely utilized as a blue colorant in the textile industry; however, its ingestion can lead to symptoms such as dizziness, diarrhea, and anaphylaxis^{3–6}. Moreover, exposure to MB can result in a range of adverse effects, including skin irritation, discoloration, and permanent eye burns, which pose risks to both human health and marine ecosystems^{3–6}. Therefore, the issue of water bodies pollution caused by the discharge of MB and other dye colorants is urgent and demands prompt action.

Various wastewater treatment methods have been employed to tackle this issue. Adsorption⁷, and coagulation/flocculation^{8,9} are among the physical techniques used to treat wastewater, whereas biological methods, such as degradation by microorganisms¹⁰ and biofilms¹¹, have also been widely researched and tested. While these methods have been proven effective, they are limited by their dye removal efficiency, dependence on contact time, and sensitivity to the presence of other compounds and/or reagents. Therefore, the advanced oxidation process, particularly in the form of photocatalysis, can be considered an effective and alternative method for the degradation of organic pollutants in water. Such a process requires a photocatalyst that transforms organic pollutants into less harmful degradation products like H₂O and CO₂, along with a light source that typically activates the photocatalyst. In most cases, natural sunlight and high-intensity light-emitting diodes (LEDs) can be used as light sources, while designing and selecting the appropriate photocatalyst remains a challenge. Ideally, the photocatalyst should be easily activated by a wide range of wavelengths, possess a large specific surface area, be made from low-cost elements, and maintain chemical stability^{12,13}. On the other hand, most metal oxide photocatalysts made from low-cost materials suffer from high electron-hole recombination rates, which reduce

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their photocatalytic efficiency. In this context, the development of a heterojunction structure can help reduce the electron-hole recombination rate, enhance the specific surface area, and potentially modify the bandgap¹⁴.

Recently, WO_3 has emerged as an effective photocatalyst for a variety of organic pollutants, ranging from textile dyes to antibiotics/pesticides^{15–17}. Typically, WO_3 is a low-cost and chemically stable n-type semiconductor with a narrow band gap of around 2.4–2.8 eV, enabling it to absorb both UV and visible light^{15–17}. On the other hand, the conduction band (CB) level of WO_3 is less negative than the potential for the single-electron reduction of oxygen. As a result, the photogenerated electrons in the CB of WO_3 cannot be effectively consumed by the adsorbed oxygen molecules to form the superoxide anion (O_2^-). This leads to the accumulation of photogenerated electrons on the surface of the WO_3 when exposed to the light, which in turn promotes the electron-hole recombination¹⁸. To tackle this issue, WO_3 can be combined with other wide bandgap materials such as ZnO, where the conduction and valence bands are properly aligned, enabling efficient charge carrier transfer. Moreover, ZnO exhibits higher electron mobility ($\sim 200\text{--}300\text{ cm}^2/\text{V}\cdot\text{s}$) compared to TiO_2 ($\sim 0.1\text{--}4.0\text{ cm}^2/\text{V}\cdot\text{s}$), making it more favorable for photocatalytic reactions¹⁹. For example, Adhikari and coworkers²⁰ synthesized and tested WO_3 -ZnO mixed oxides for the photocatalytic degradation of cationic MB and anionic Orange G dyes. Experimental results revealed that the 10% WO_3 -containing ZnO exhibited better photocatalytic performance than the other compositions for both dyes. Sajjad and coworkers examined the impact of ZnO incorporation (1–4 wt%) into WO_3 to create ZnO/ WO_3 composites and evaluated their photocatalytic performance using methyl orange dye²¹. It was found that 2 wt% ZnO/ WO_3 sample demonstrated the highest photocatalytic performance under visible light illumination. In general, the efficiency of the system is governed by multiple parameters, including the synthesis methodology, catalyst size and morphology, specific surface area and porosity, the ZnO/ WO_3 ratio, etc.

In our study, a hydrothermal treatment was employed to facilitate the deposition of ZnO onto the surface of the WO_3 structure, which differs from the methods reported before^{20,21}, resulting in a composite that combines enhanced crystallinity, increased pore volume, and improved charge separation without the need for noble metals or complex surface modifications. Several ZnO/ WO_3 ratios were evaluated based on their photocatalytic performance using MB as a model pollutant under simulated solar light for optimization purposes. The use of simulated solar light, compared to UV irradiation, enhances the applicability of photocatalysts under natural sunlight, thereby improving energy efficiency and environmental sustainability.

Materials and methods

Materials and photocatalyst synthesis

High-purity reagents were procured from Merck and used without further purification. WO_3 was synthesized via a precipitation-calcination method. In brief, 0.75 g of $\text{Na}_2\text{WO}_4\cdot 2\text{H}_2\text{O}$ was dispersed in 10 mL of deionized water, and the pH of the solution was adjusted to 1 using 1 M HCl. The mixture was stirred for 2 h, then centrifuged at 7000 rpm for 5 min. The resulting precipitate was washed with deionized water and ethanol, dried at 60 °C overnight, and subsequently calcined at 500 °C for 2 h.

The synthesis of ZnO/ WO_3 -based catalysts was performed using a hydrothermal method. In this procedure, equimolar amounts of $\text{Zn}(\text{NO}_3)_2\cdot 6\text{H}_2\text{O}$ and $\text{C}_6\text{H}_{12}\text{N}_4$ (hexamethylenetetramine, HMTA) were added. The amount of the zinc precursor was controlled at 5%, 10%, and 25% by weight relative to the mass of WO_3 . The reagents were added to 20 mL of deionized (DI) water and sonicated for 1 min to ensure homogeneity. The suspension was then transferred to a Teflon-lined autoclave and subjected to hydrothermal treatment at 95 °C for 12 h. After natural cooling, the precipitate was separated by centrifugation, washed several times with DI water and ethanol, and dried overnight at 60 °C.

Catalyst characterization

The surface morphology of the catalysts was examined using a Crossbeam 540 (Carl Zeiss, Germany) scanning electron microscope (SEM) equipped with energy-dispersive X-ray (EDX). In addition, samples were characterized using a JEM2010F transmission electron microscope (TEM, JEOL Ltd, Japan). The crystallographic analysis of the prepared catalysts was evaluated using a SmartLab X-ray diffractometer (XRD, Rigaku Corp., Japan). The surface area and pore size distribution were determined using an Autosorb iQ nitrogen porosimeter (Quantachrome Instruments, USA). Elemental analysis was performed employing an Inductively Coupled Plasma Optical Emission Spectroscopy (ICP-OES, iCap 6000, Thermo Fischer Scientific, USA). Light absorbance measurements were carried out using an absolute quantum yield spectrometer equipped with an integrated camera (C9920-02, Hamamatsu Photonics, Japan).

Photocatalytic activity measurements

Prior to the photocatalytic treatment, the MB solution with the catalyst was kept in the dark for 30 min to establish an adsorption-desorption equilibrium. The photocatalytic efficiency of the synthesized catalysts was assessed using an LCS-100 solar simulator (100 W, AM 1.5G filter, Newport-Spectra Physics, GmbH, Germany) under controlled conditions: MB concentration of 5 mg/L, MB solution volume of 30 mL, catalyst dosage of 100 mg/L, and 5 μL of H_2O_2 (35%). The light intensity was calibrated to 1 sun using a silicon reference cell. After a predetermined reaction time, the solution was centrifuged at 7000 rpm for 4 min, and the absorbance of the MB solution was measured at 554 nm using a Genesys 50 UV-Vis spectrophotometer (Thermo Fisher Scientific Inc., USA). All experiments were conducted in triplicate to ensure reproducibility.

To investigate the photocatalytic degradation mechanism, radical scavenger experiments were conducted using 5 mM of each scavenger²². Ethylene diamine tetra-acetate (EDTA) was used to quench photogenerated holes (h^+), isopropyl alcohol (IPA) was employed as a hydroxyl radical scavenger, and benzoquinone (p-BQ) was used to capture superoxide radicals. The scavengers were added individually to the reaction solution prior to

irradiation, and the degradation of MB was monitored under the same experimental conditions as the standard photocatalytic tests.

Results and discussion

Samples characterization

The morphology of the prepared samples was analyzed using SEM. Figure 1a illustrates that the bare WO_3 exhibits a typical plate-like structure with a relatively smooth surface morphology. One can notice that with increasing ZnO content, the particles exhibit a rougher and more porous structure, which could potentially enhance the photocatalytic performance of the WO_3 -based catalysts by increasing their specific surface area (Figs. 1b–d). Pure ZnO (Fig. 1e) consists of hexagonal-shaped, rod-like structures, which are commonly formed as a result of the anisotropic growth of ZnO^{23,24}. Elemental mapping of 5% ZnO/ WO_3 (Fig. 1f) reveals a uniform distribution of ZnO across the surface of WO_3 . The actual ZnO content on the WO_3 surface was quantitatively determined using ICP-OES and presented in Table S1 (Supporting Information). It can be observed that the actual concentrations were slightly lower than the theoretical values, which can be mainly attributed to losses of the Zn precursor during the synthesis process or detachment of ZnO during the purification process.

Next, XRD analysis (Figure S1, Supporting Information) was performed to assess the crystallographic structure of the prepared catalysts. Bare WO_3 exhibited prominent XRD peaks assigned to the (002), (020), (200), and (112) planes, which are consistent with its monoclinic crystalline structure^{25,26}. Following the incorporation of 5% ZnO, the catalyst retained the characteristic XRD pattern of WO_3 , suggesting that the WO_3 crystal structure remains dominant. Among all the catalysts, 5% ZnO/ WO_3 exhibited the highest XRD peak intensities, indicating better crystallinity. Although the primary WO_3 peaks are still present in the XRD pattern of 10% ZnO/ WO_3 , their decreased intensity indicates that the ZnO phase becomes more prominent at higher ZnO concentrations. At 25% ZnO, the WO_3 diffraction peaks become less distinct, suggesting that the stronger crystallinity and higher proportion of ZnO may be masking the presence of the WO_3 phase. The diffraction profile of pure ZnO reveals well-defined peaks corresponding to its hexagonal wurtzite structure²⁷. Notably, the intense peaks at 31.8° (100), 34.4° (002), and 36.2° (101) confirm that the synthesized ZnO is highly crystalline and exhibits phase purity, with no detectable secondary phases²⁸.

The average crystallite size (D) of the prepared photocatalysts was calculated from XRD data using the well-known Debye-Scherrer equation. Based on the calculated results (Table S2, Supporting Information), bare ZnO exhibits the highest crystallite size of 41.36 nm, while 5% ZnO/ WO_3 has the smallest one ~18.05 nm. The crystallite size of bare WO_3 is found to be 22.88 nm, suggesting that the presence of a small ZnO amount (5%) hinders the crystallite growth of WO_3 . With a higher ZnO percentage (10% and 25%), the crystallite size of ZnO/ WO_3 composites increases, potentially indicating a stabilization effect.

Dislocation density represents the number of defects within the sample, defined as the total length of dislocation lines per unit volume of the crystal, and is calculated using Eq. (1)²⁹.

$$\delta = \frac{1}{D^2} \quad (1)$$

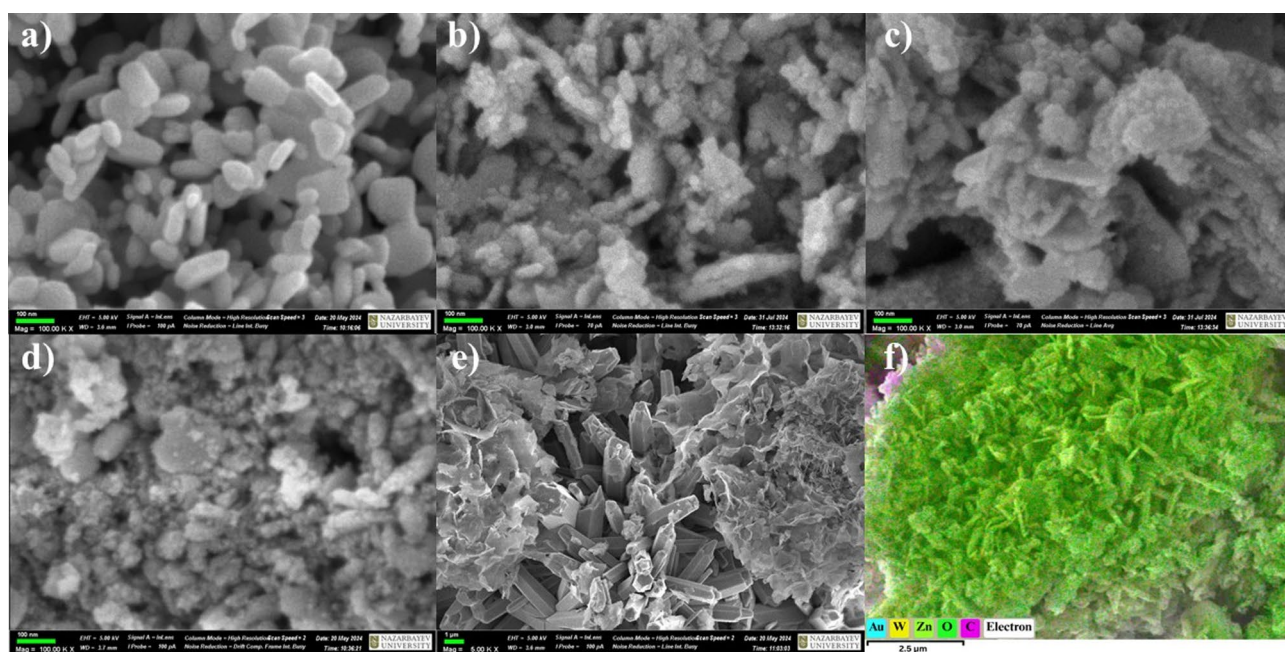


Fig. 1. SEM images of the synthesized catalysts: (a) WO_3 , (b) 5% ZnO/ WO_3 , (c) 10% ZnO/ WO_3 , (d) 25% ZnO/ WO_3 , (e) ZnO, and EDS analysis of (f) 5% ZnO/ WO_3 . Note: The scale bars are 100 nm for (a–d) and 1 μm for (e) due to different particle sizes.

Catalysts	BET surface area, m ² /g	Pore volume, cm ³ /g
WO ₃	16.02	0.080
5% ZnO/WO ₃	21	0.139
10% ZnO/WO ₃	22.42	0.106
25% ZnO/WO ₃	25.58	0.132
ZnO	25	0.114

Table 1. BET/BJH analysis data.

Catalysts	Direct bandgap, eV	Indirect bandgap, eV
WO ₃	2.61	2.54
5% ZnO/WO ₃	2.67	2.60
10% ZnO/WO ₃	2.68	2.61
25% ZnO/WO ₃	2.68	2.63
ZnO	3.12	3.06

Table 2. The bandgap energy values for the prepared structures.

The dislocation density (δ) exhibits an anticipated inverse relationship with crystallite size: bare ZnO shows the lowest dislocation density, whereas the 5% ZnO/WO₃ composite shows the highest dislocation density, indicating a greater concentration of structural defects (Table S2, Supporting Information). For bare WO₃ and other ZnO/WO₃ composites, the dislocation density remains relatively consistent but decreases slightly as the ZnO content increases, possibly due to structural stabilization.

The lattice strain (ϵ) of the photocatalysts was calculated using Eq. (2) (Stokes-Wilson equation)²⁹:

$$\epsilon = \beta \cos\theta / 4 \quad (2)$$

The highest lattice strain is observed in 5% ZnO/WO₃ composite, aligning well with the higher dislocation density and reduced crystallite size (Table S2, Supporting Information). The composites containing 10% and 25% ZnO exhibit slightly higher strain values compared to bare WO₃, indicating that ZnO incorporation induces lattice distortion; however, the strain appears to stabilize with increasing ZnO content, suggesting a potential structural accommodation of ZnO.

Next, the specific surface area and pore volume of the catalysts were evaluated using the Brunauer-Emmett-Teller (BET) and Barrett-Joyner-Halenda (BJH) analyses derived from nitrogen adsorption-desorption isotherms. The results of the porosimetry analysis are presented in Figure S2 (Supporting Information) and Table 1. According to the IUPAC classification, the N₂ adsorption-desorption isotherms of the catalysts correspond to type IV with H3 hysteresis loops, confirming the presence of mesoporous structures³⁰. As shown in Table 1, incorporation of ZnO resulted in an increase in both the BET-specific surface area and pore volume of the catalysts. Although the specific surface areas of 10% ZnO/WO₃, 25% ZnO/WO₃, and even ZnO were higher, the 5% ZnO/WO₃ catalyst exhibited the largest pore volume among all samples. In addition, the 5% ZnO/WO₃ catalyst exhibited higher dislocation density and lattice strain, which can synergistically govern its photocatalytic activity.

The band gaps of all prepared structures were estimated using the Tauc equation, with the graphs presented in Figure S3 (Supporting Information) and values summarized in Table 2.

While WO₃ is generally classified as an indirect bandgap semiconductor, both direct and indirect transitions are frequently reported, reflecting the influence of crystal phase, morphology, and related structural parameters on its electronic structure. For example, bare WO₃ displayed bandgap values of 2.61 eV (direct) and 2.54 eV (indirect), which align well with those reported in the literature^{31,32}. With a direct bandgap of 3.12 eV, the ZnO catalyst showed a higher value than all WO₃-based photocatalysts, but in good agreement with previously reported data³³. It can be observed that both the direct and indirect band gaps of ZnO/WO₃ composites increase slightly with ZnO incorporation; however, these changes do not substantially alter their light absorption properties.

Next, the surface chemical composition and oxidation states of the 5% ZnO/WO₃ composite were examined using XPS (Fig. 2).

The survey spectrum confirms the presence of W, Zn, O, and C elements, with no additional impurity signals detected, demonstrating the high purity of the synthesized composite. The carbon peak originates from adventitious surface hydrocarbons and was used for charge calibration. The high-resolution W 4f spectrum can be deconvoluted into two characteristic peaks located at binding energies of 37.6 eV (W 4f_{5/2}) and 35.5 eV (W 4f_{7/2}). The ~2.1 eV spin-orbit separation corresponds well to reported values for W⁶⁺ species, confirming that tungsten exists predominantly in the 6+ oxidation state within the WO₃ lattice³⁴. The Zn 2p core-level spectrum exhibits two spin-orbit components centered at 1044.4 eV (Zn 2p_{1/2}) and 1021.3 eV (Zn 2p_{3/2}). These values match well with literature-reported positions for Zn²⁺ in ZnO, indicating that Zn is present in its typical 2+ oxidation state in the composite³⁵. The O 1s spectrum can be fitted into two peaks at 530.1 eV and 531.7 eV.

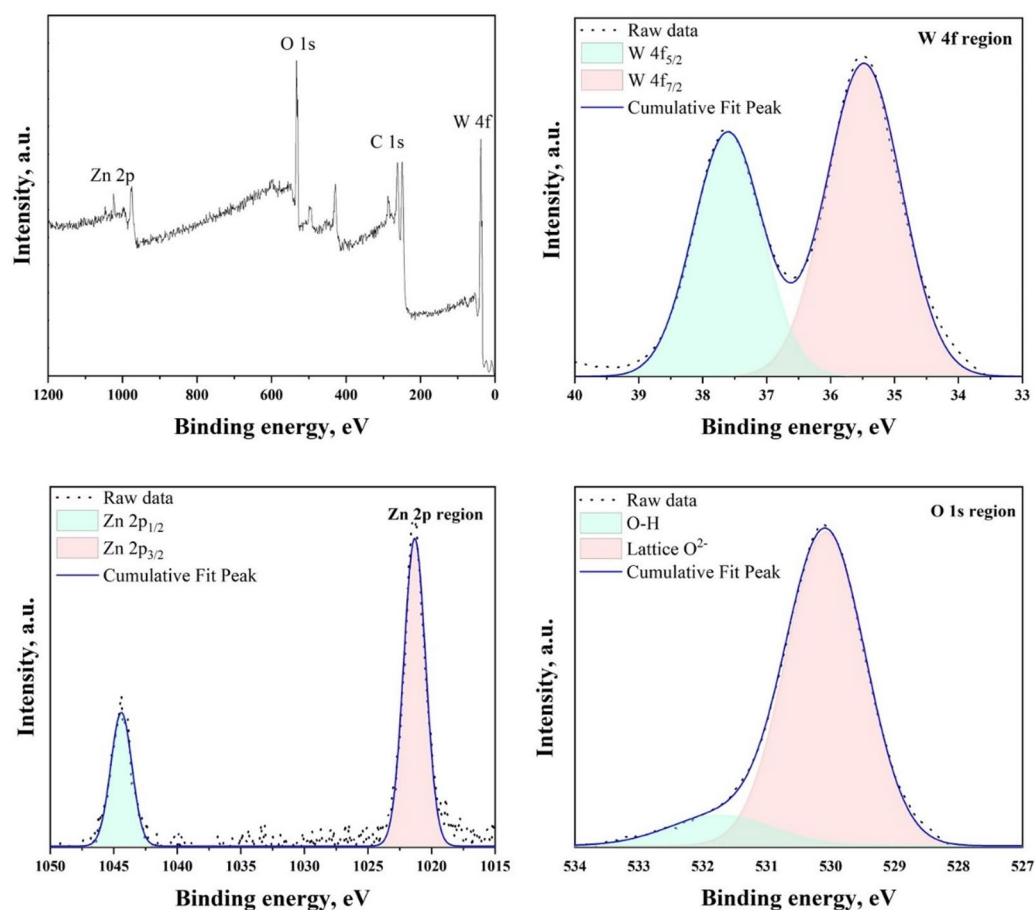


Fig. 2. XPS survey and high-resolution spectra of the 5% ZnO/WO₃ composite.

The dominant peak at 530.1 eV is attributed to lattice oxygen (O²⁻) in WO₃ and ZnO, while the higher-binding-energy component at 531.7 eV corresponds to surface hydroxyl groups (-OH) and adsorbed oxygen species. The presence of these surface oxygen species is beneficial for photocatalysis, as they can serve as active sites for radical generation. Overall, the XPS results confirm the successful formation of the ZnO/WO₃ composite containing W⁶⁺ and Zn²⁺ species, with abundant surface oxygen functionalities that can support photocatalytic activity.

Photocatalytic degradation of methylene blue

The photocatalytic activity of the ZnO/WO₃ structures was assessed through the degradation of MB dye under simulated solar irradiation (Fig. 3). The tests were conducted with reaction times of 30 and 60 min, an initial MB concentration of 5 mg/L (volume = 30 mL), a catalyst dosage of 100 mg/L, and the addition of 5 μ L of H₂O₂ (35%). It is clear that bare WO₃ and ZnO showed moderate photocatalytic performance, whereas the ZnO/WO₃-based composites demonstrated enhanced activity under simulated solar light. In particular, 5% ZnO/WO₃ samples achieved 93.8% MB degradation after 60 min, representing a 21.2% improvement over bare WO₃ and the highest photocatalytic activity among all the samples. After 60 min, the 10% ZnO/WO₃ and 25% ZnO/WO₃ composites achieved MB degradation efficiencies of 82.3% and 80.4%, respectively, while pure ZnO showed the lowest activity (~68.5%). These results indicate that 5% ZnO/WO₃ composite provides the highest photocatalytic efficiency, likely due to an optimal ZnO-WO₃ ratio that improves the light absorption and charge separation. Hence, 5% ZnO/WO₃ composite was selected for further in-depth evaluation of its photocatalytic activity under different experimental conditions.

A more detailed kinetic study of MB degradation can be observed in Fig. 4. Since H₂O₂ is also capable of degrading organic pollutants, the photocatalytic degradation of MB was studied under three conditions: MB only, MB + H₂O₂, and MB + H₂O₂ + 5% ZnO/WO₃ photocatalyst.

Figure 4a demonstrates that MB undergoes minimal self-degradation under simulated solar irradiation, with only ~10% degradation observed after 60 min. The addition of H₂O₂ had a more pronounced effect, enhancing MB degradation up to 55%. In comparison, the introduction of the 5% ZnO/WO₃ photocatalyst substantially enhanced the reaction rate, resulting in ~93.8% MB degradation after 60 min. Figure 4b compares the kinetics of each degradation method. In the presence of 5% ZnO/WO₃ photocatalyst, the apparent rate constant for MB degradation under simulated solar light reached 0.02054 min⁻¹, representing ~3.24-fold and ~9.13-fold enhancements relative to the MB + H₂O₂ and MB-only systems, respectively. For reference, a comparative

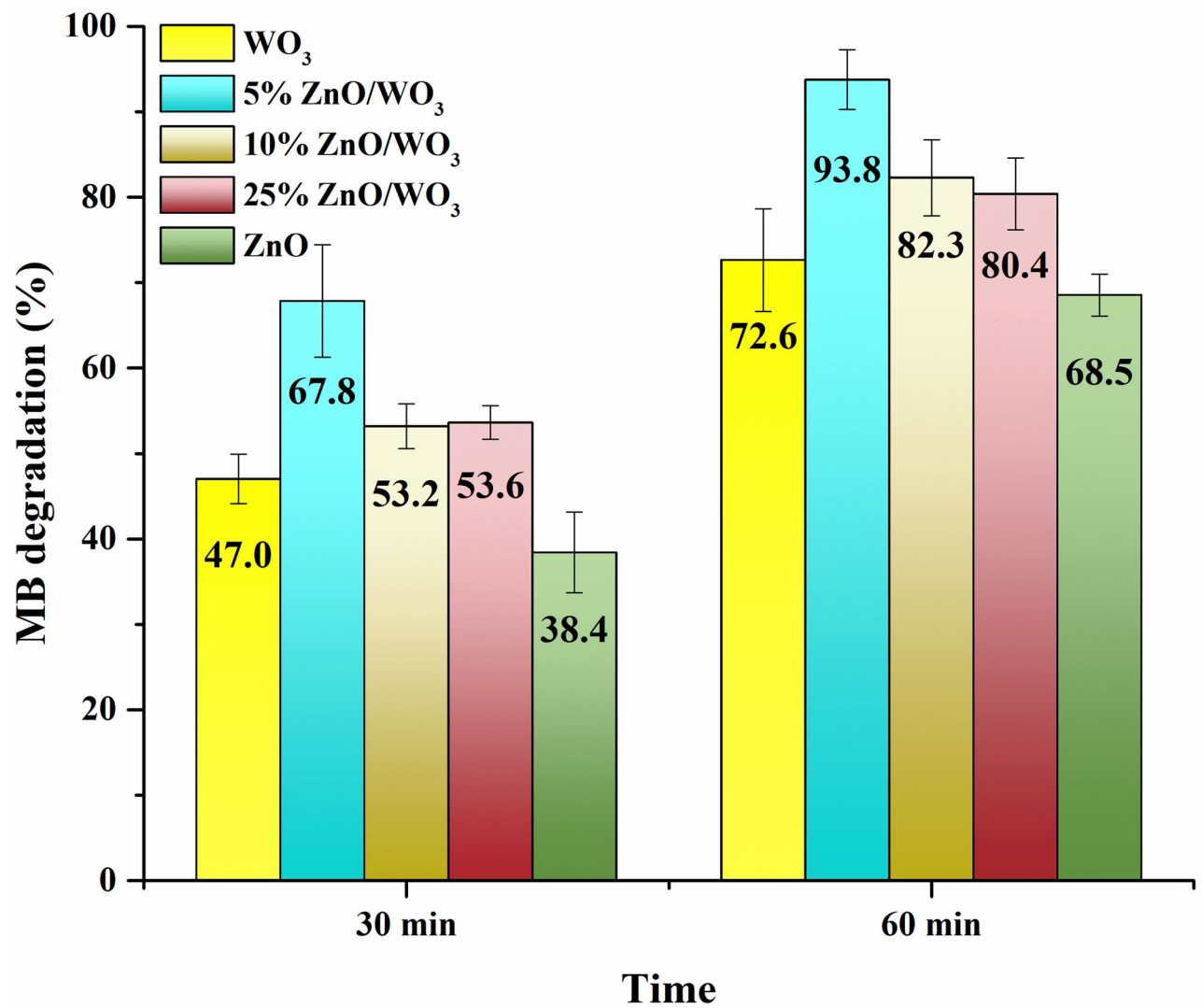


Fig. 3. Degradation of MB dye (5 mg/L) under simulated solar light irradiation.

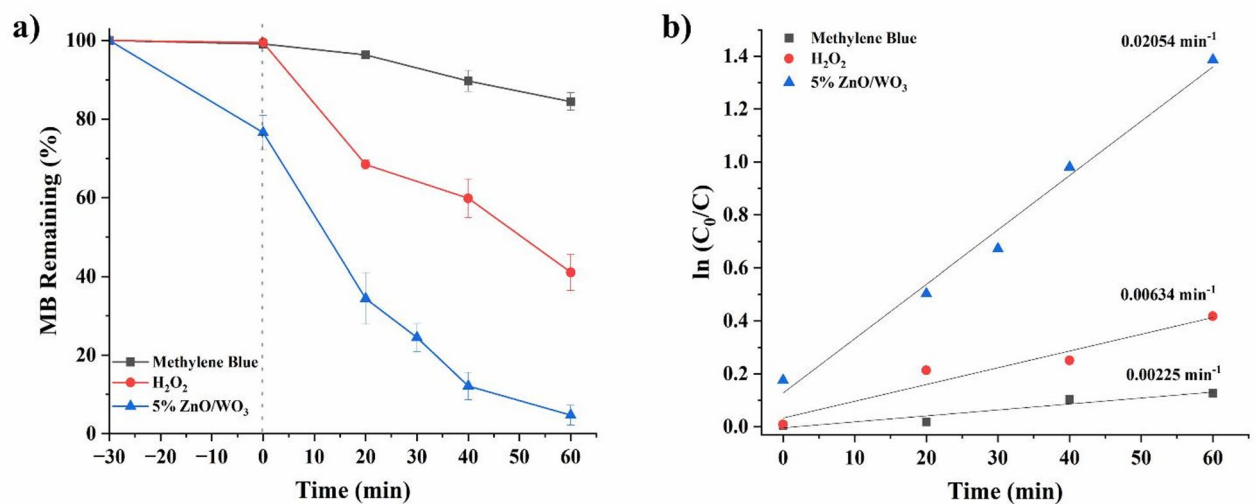


Fig. 4. Photodegradation of MB versus time (a) and kinetics (b).

Catalyst type	Catalyst, mg/L	Light source	H ₂ O ₂ , mL/L	MB, mg/L	MB deg., %	Time, min	k, min ⁻¹	Refs.
Monoclinic-WO ₃	500	Visible	0	40	96	120	-	36
1%rGO-WO ₃	300	Visible	0	10	65	60	0.01129	37
MnO _x /WO ₃	200	-	200 (0.1 M)	10	95	60	-	38
Pt/cubic-WO ₃	1000	Visible	0	3.2	67	60	0.0531	39
WO ₃	300	Solar	0.167 (35%)*	4	98	40	-	40
WO ₃ -GO	500	Visible	0	3.2	82	70	-	41
Pt/WO ₃ -GO	500	Visible	0	3.2	94	70	-	41
5%ZnO/WO ₃	100	Solar	0.167 (35%)	5	93.8	60	0.02054	This work

Table 3. Comparison of WO₃-based photocatalytic systems for MB degradation. *One drop of H₂O₂.

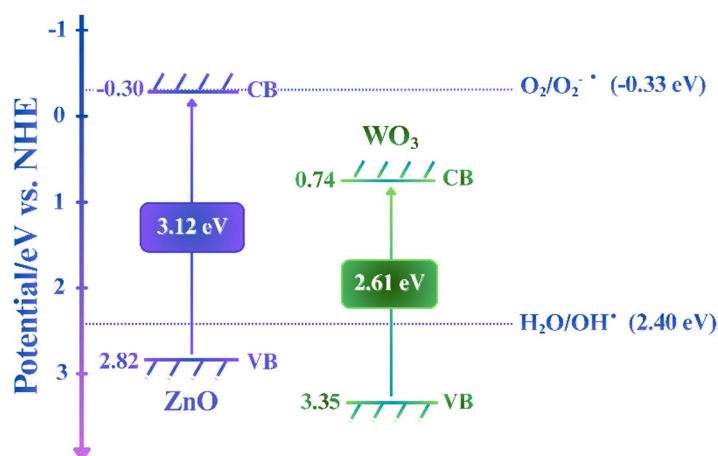


Fig. 5. Estimated energy band positions of ZnO and WO₃.

analysis with other WO₃-based photocatalysts reported for MB degradation was carried out, and the outcomes are presented in Table 3. Several parameters were taken into account, including catalyst type/dosage, irradiation source and duration, pollutant and oxidant concentrations, as well as degradation percentage and reaction rate.

According to Table 3, the 5% ZnO/WO₃ photocatalyst exhibited a high degradation efficiency of MB (93.8%) at a relatively low dosage of 100 mg/L, which is below the catalyst loadings employed in all other studies. Unlike most studies that used visible light, this work employed solar irradiation, which is preferable for future practical applications. With respect to photocatalytic activity, the 5% ZnO/WO₃ photocatalyst achieved a notable apparent rate constant of 0.02054 min⁻¹ without relying on costly metals or complex surface modifications. Hence, it can be inferred that the 5% ZnO/WO₃ photocatalyst offers an optimal balance between low catalyst dosage and high MB degradation relative to other WO₃-based systems.

ZnO and WO₃ are both n-type semiconductors, exhibiting direct bandgap energies of 3.12 eV and 2.61 eV, respectively. Using Mulliken electronegativities (X) and the empirical relations, the valence- and conduction-band potentials (Eqs. 3 and 4) were calculated⁴²:

$$E_{VB} = X - E_e + 0.5E_g \quad (3)$$

$$E_{CB} = E_{VB} - E_g \quad (4)$$

For WO₃ (X = 6.54 eV)⁴², the VB and CB edge positions are +3.35 eV and +0.74 eV vs. NHE, respectively. For ZnO (X = 5.76 eV)⁴², the corresponding VB and CB edges are +2.82 eV and -0.30 eV. Based on these band positions, the ZnO/WO₃ composite forms a type-II staggered heterojunction. The slight bandgap narrowing observed for the 5% ZnO/WO₃ composite (2.67 eV) further supports enhanced interfacial charge interaction between the two oxides.

Figures 5 and 6 and the corresponding Eqs. (5–9) present the proposed energy band positions and pathway for MB degradation via ZnO/WO₃ photocatalysis in the presence of H₂O₂. Upon light activation, electrons and holes will be formed in both semiconductors (Eq. 5). Since the Fermi level of ZnO is higher than that of WO₃, electrons preferentially transfer from ZnO to WO₃, rendering WO₃ an effective electron sink in the coupled oxide system^{43,44}. The reduction of O₂ by electrons in the CB of WO₃ is thermodynamically unfavorable as the CB edge potential of WO₃ is more positive than the standard redox potential of the O₂/O₂•⁻ couple⁴⁵. Thus, accumulated electrons can react with hydrogen peroxide to produce one hydroxyl radical and one hydroxide ion (Eq. 6). At the same time, the photogenerated holes can be transferred from the valence band (VB) of WO₃ to the VB of ZnO. Photogenerated holes can either directly oxidize MB or generate highly reactive hydroxyl radicals

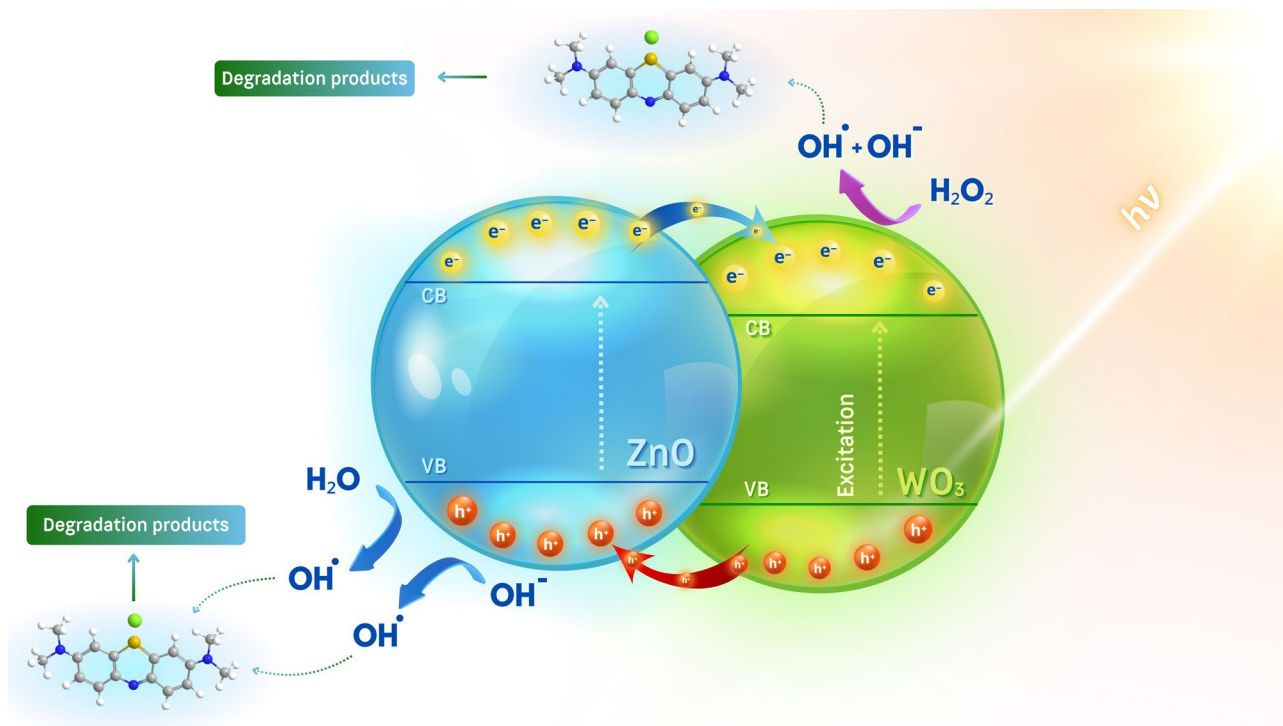
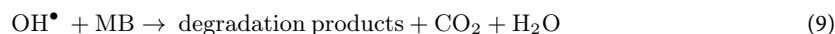
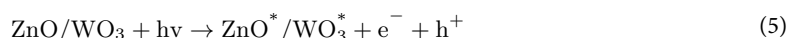


Fig. 6. Proposed mechanism for MB degradation using ZnO/WO₃ photocatalyst.

using hydroxide ions or water molecules (Eqs. 7, 8)^{46,47}. Radical scavenger experiments (Figure S4) confirmed the dominant role of hydroxyl radicals in MB degradation: in the absence of scavengers, MB degradation reached 93.8%, whereas in the presence of IPA (OH• scavenger), it decreased to 75.8%, indicating a significant contribution of hydroxyl radicals. The addition of EDTA (h• scavenger) and p-BQ (O₂•⁻ scavenger) slightly reduced the degradation efficiency to 89.2% and 86.7%, respectively, suggesting that holes and superoxide radicals also participate in the degradation process, but to a lesser extent. Finally, MB molecules are mineralized by producing CO₂, H₂O, and some degradation by-products (Eq. 9).



The influence of pH on MB degradation was also examined after 60 min of photocatalytic reaction (Figure S5a, Supporting Information). At acidic conditions (pH 4.4), MB degradation reached 81%, whereas complete degradation was achieved under alkaline conditions (pH 8.5). This enhancement can be attributed to the higher concentration of hydroxyl ions in basic media, which are readily oxidized by photogenerated holes to form highly reactive OH• radicals (Eq. 7). Moreover, under alkaline conditions, the catalyst surface acquires a negative charge, while MB is a cationic dye, leading to enhanced dye adsorption through the electrostatic interactions⁴⁸. In contrast, under acidic conditions, ZnO component can be easily etched or ZnO/WO₃ photocatalyst acquires a positive surface charge, which can repel the dye molecules. As shown in Figure S5b (Supporting Information), a catalyst dosage of 33.3 mg/L resulted in 87.6% MB degradation, whereas increasing the dosage to 166.7 mg/L enhanced the degradation efficiency up to 95.9%. Typically, regulation of the photocatalyst concentration provides control over degradation efficiency, but very high loadings can have an adverse effect due to the turbidity of the solution and reduced light penetration⁴⁹. Furthermore, employing excessively high catalyst dosages is not economically feasible; therefore, adjusting the solution pH or extending the treatment duration provides a more practical alternative, as noted earlier. The reusability tests over five consecutive runs revealed a marked loss of activity, with degradation efficiency decreasing from 93.8% initially to 81.0% and 64.2% in the second and third cycles, respectively, followed by 62.8% and 59.9% in the fourth and fifth cycles (Figure S5c, Supporting Information). The XPS analysis of the recycled ZnO/WO₃ composite showed no significant shifts in the binding energies of the W 4f, Zn 2p, or O 1s core levels, with peaks appearing at 35.7 and 37.9 eV (W⁶⁺), 1021.3 and 1044.4 eV (Zn²⁺), and 530.0 and 532.5 eV for lattice and surface oxygen species, respectively (Figure S6, Supporting Information). These values closely match those of the fresh photocatalyst, indicating that the

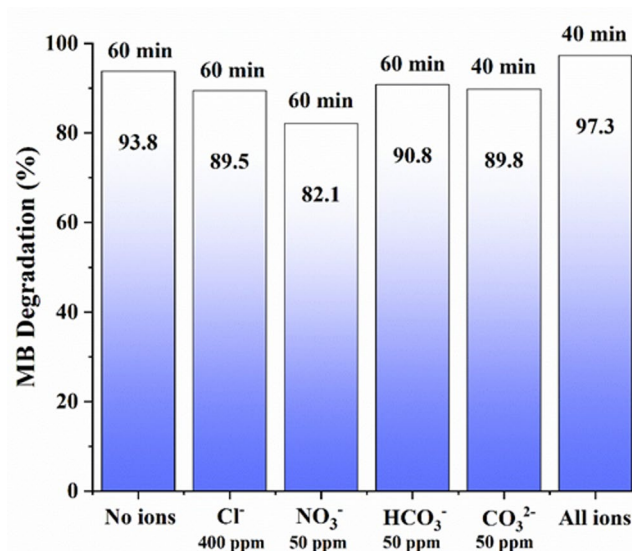


Fig. 7. MB degradation with 5% ZnO/WO₃ in the presence of different ions.

chemical states of W, Zn, and O remain preserved after repeated photocatalytic cycles. Therefore, a reduction in performance is likely attributable to catalyst photocorrosion or surface fouling by reaction intermediates not detectable by XPS, highlighting its limited durability during prolonged operation⁵⁰. This stability issue can be resolved by depositing a more stable thin TiO₂ layer on the surface of the photocatalyst, which will be addressed in future work.

Figure 7 shows the effect of different anions on the photocatalytic degradation efficiency of MB by 5% ZnO/WO₃, with the concentrations selected to mimic those occurring in wastewater²⁶. In the presence of 400 ppm Cl⁻, the degradation efficiency reached 89% under the same conditions, indicating a moderate inhibitory influence of Cl⁻ ions on the photocatalytic process. By contrast, NO₃⁻ ions (50 ppm) exerted a slightly stronger suppression, reducing the efficiency to 82%. Unlike Cl⁻ and NO₃⁻, bicarbonate ions (HCO₃⁻, 50 ppm) showed a minimal inhibitory influence, with 91% of MB degradation achieved after 1 h. Notably, the addition of 50 ppm CO₃²⁻ led to 90% degradation, whereas the mixture of all anions enhanced the efficiency to 97% with a shortened reaction time of 40 min. Carbonate ions are common constituents of natural waters, and can interact with hydroxyl radicals to yield carbonate radicals according to Eq. (10)⁵¹.



Although hydroxyl radicals possess a higher redox potential than carbonate radicals, the latter can still promote MB degradation owing to their extended lifespan and high selectivity^{27,51}. In summary, anions introduced at near-natural or elevated concentrations showed no significant impact on the photocatalytic degradation process, suggesting that subsequent studies may be carried out with actual textile wastewater.

Conclusions

This study demonstrates that the synthesized ZnO/WO₃ composite structure achieves markedly improved photocatalytic activity toward MB degradation under simulated solar irradiation, outperforming bare ZnO and WO₃ structures. Notably, the 5% ZnO/WO₃ composite was found to be the most effective, offering larger pore volume, improved charge carrier separation, and enhanced photocatalytic performance in MB dye degradation. A comparative analysis with other WO₃-based photocatalysts suggested that the 5% ZnO/WO₃ composite is more effective in terms of cost, catalyst dosage, light source, degradation time, and apparent rate constant. Furthermore, assessment of photocatalytic performance in the presence of anions typically found in wastewater confirms that the proposed catalyst can operate effectively under realistic conditions. Mechanistic investigations provided additional insight into the photocatalytic behavior of the composite. Radical scavenger experiments revealed that hydroxyl radicals are the dominant reactive species responsible for MB degradation, while photogenerated holes and superoxide radicals contribute to a lesser extent. The XPS analysis confirmed that the oxidation states of W⁶⁺, Zn²⁺, and lattice oxygen remained unchanged after 5 photocatalytic cycles, indicating that the bulk chemical structure is preserved during operation. While reusability tests indicated certain stability concerns due to the possible partial photocorrosion of ZnO or surface fouling by degradation intermediates, the proposed 5% ZnO/WO₃ composite still represents a viable option for the efficient degradation of organic pollutants. Future improvements, such as coating with a thin and stable TiO₂ layer, may help overcome these limitations and will be addressed in future studies.

Data availability

The data supporting this article have been included as part of the Supporting Information.

Received: 18 September 2025; Accepted: 11 February 2026

Published online: 18 February 2026

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Author contributions

This manuscript was written through the contributions of all authors. All authors approved the final version of the manuscript.

Funding

This research has been funded by the Committee of Science of the Ministry of Science and Higher Education of the Republic of Kazakhstan (Grant No. BR28712489). This research was also funded by Nazarbayev University FDCRDG (Grant No. 20122022FD4111).

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1038/s41598-026-40207-0>.

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SUPPORTING INFORMATION

ZnO/WO₃ composite for efficient photocatalytic degradation of Methylene blue dye under solar light

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Table S1. ZnO content in ZnO/WO₃ structures determined by ICP-OES.

Catalysts	ZnO, %
5% ZnO/WO ₃	3.76
10% ZnO/WO ₃	7.27
25% ZnO/WO ₃	20.56

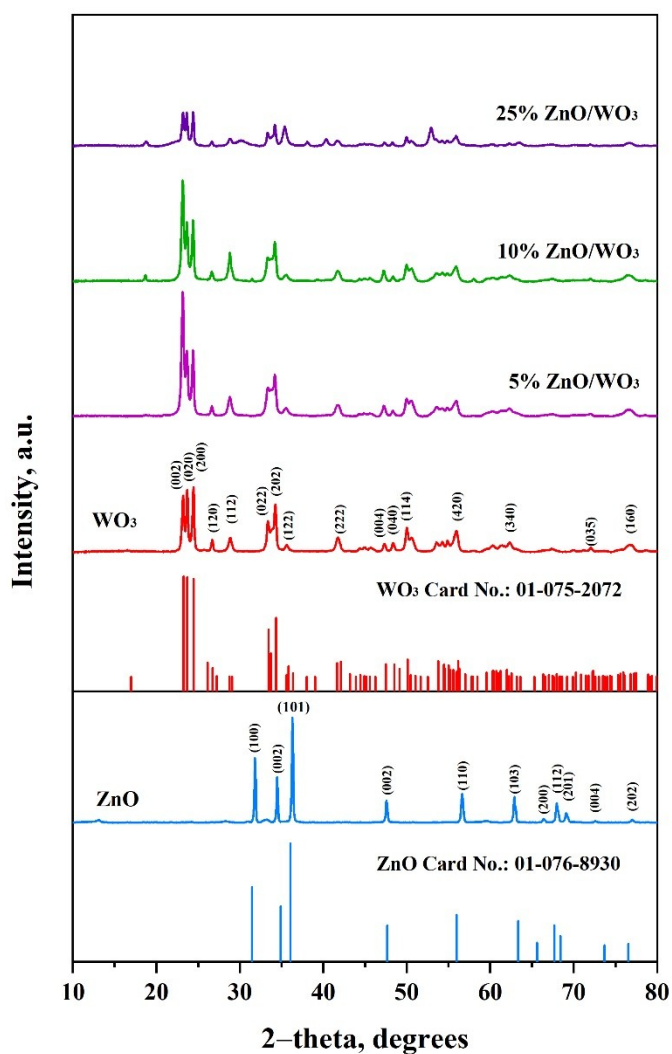


Figure S1. XRD patterns of bare WO₃, ZnO, and synthesized ZnO/WO₃ composites.

Table S2. Average crystallite size, dislocation density, and lattice strain of the prepared structures.

Catalysts	Crystallite size, nm	$\delta \times 10^{-3} \text{ (nm}^{-2}\text{)}$	$\epsilon \times 10^{-3}$
WO ₃	22.88	4.97	4.97
5% ZnO/WO ₃	18.05	8.4	6.67
10% ZnO/WO ₃	23.63	4.96	5.42
25% ZnO/WO ₃	21.32	4.92	5.81
ZnO	41.36	2.72	5.02

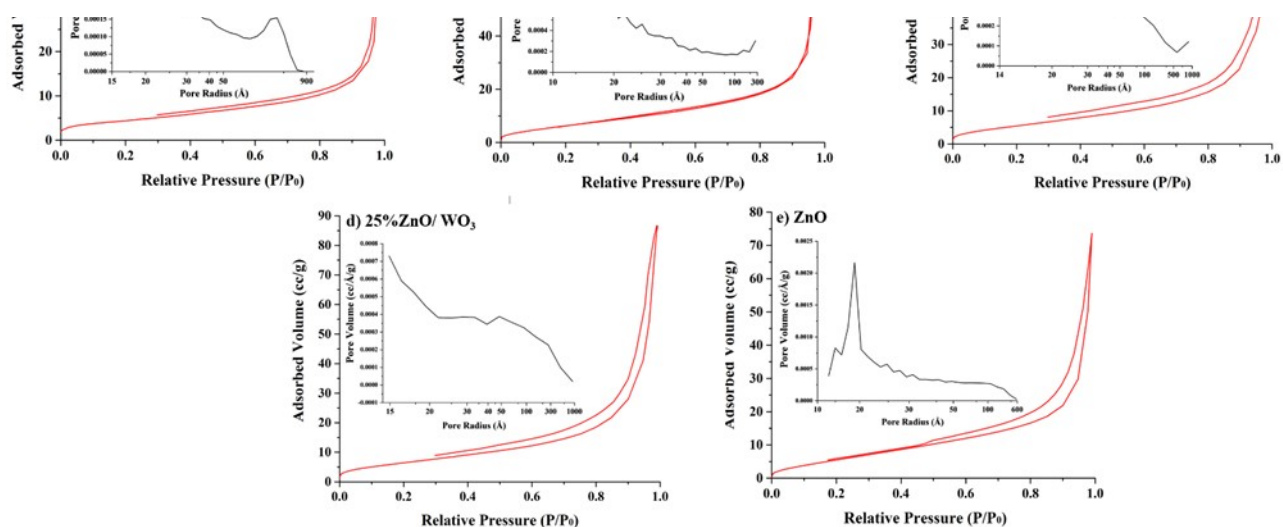
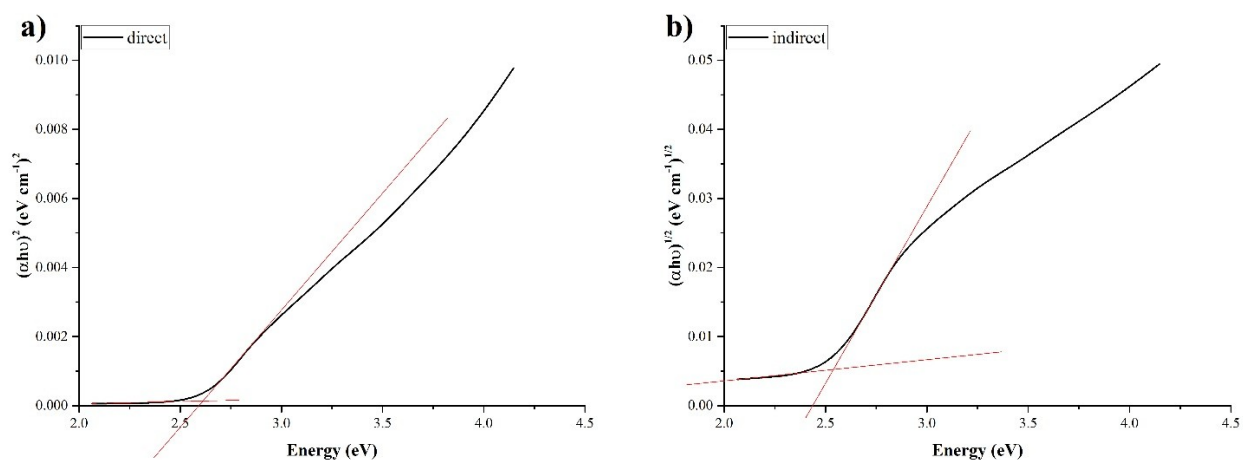
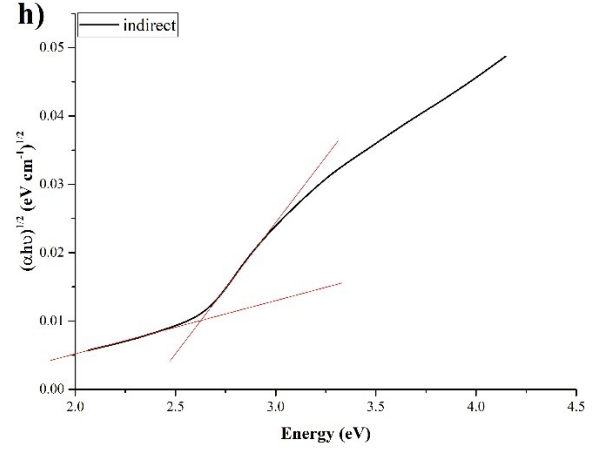
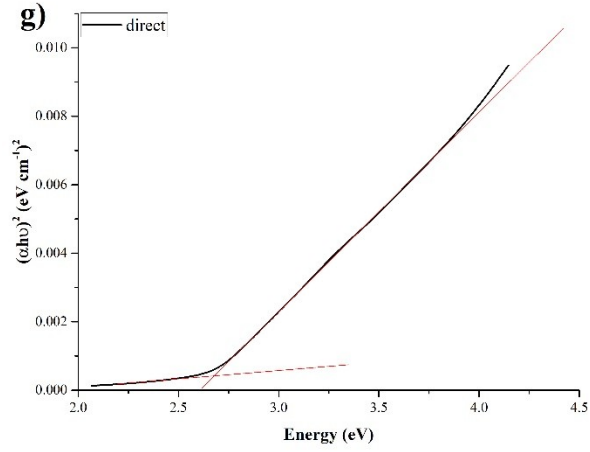
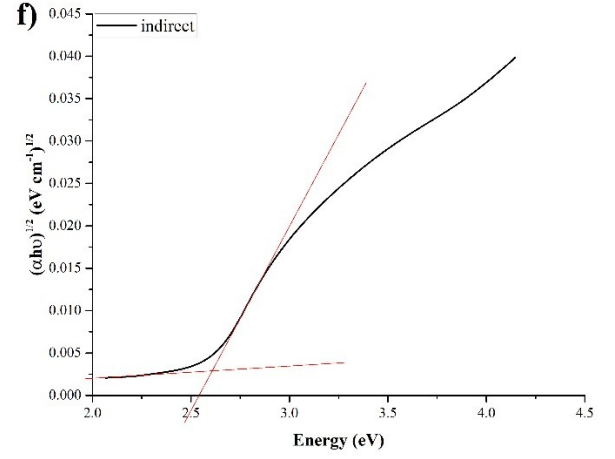
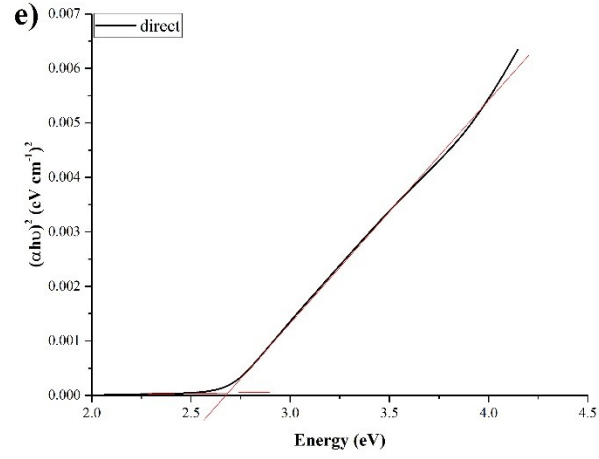
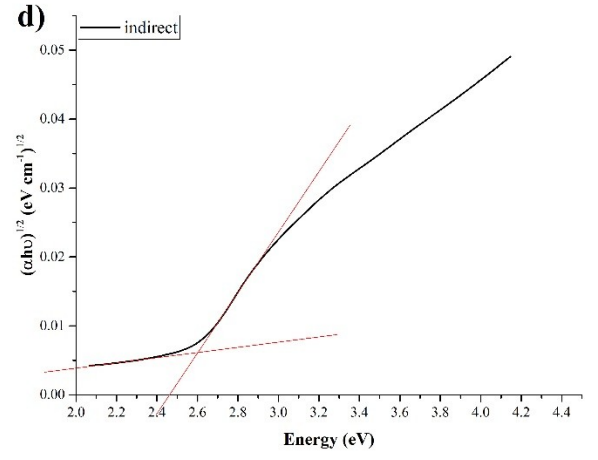
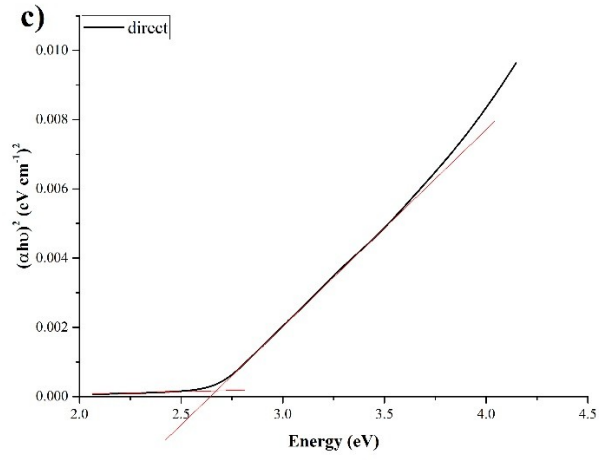


Figure S2. Nitrogen adsorption-desorption isotherms and pore size distribution data (inset) for the bare WO_3 , ZnO/WO_3 heterojunctions, and bare ZnO .





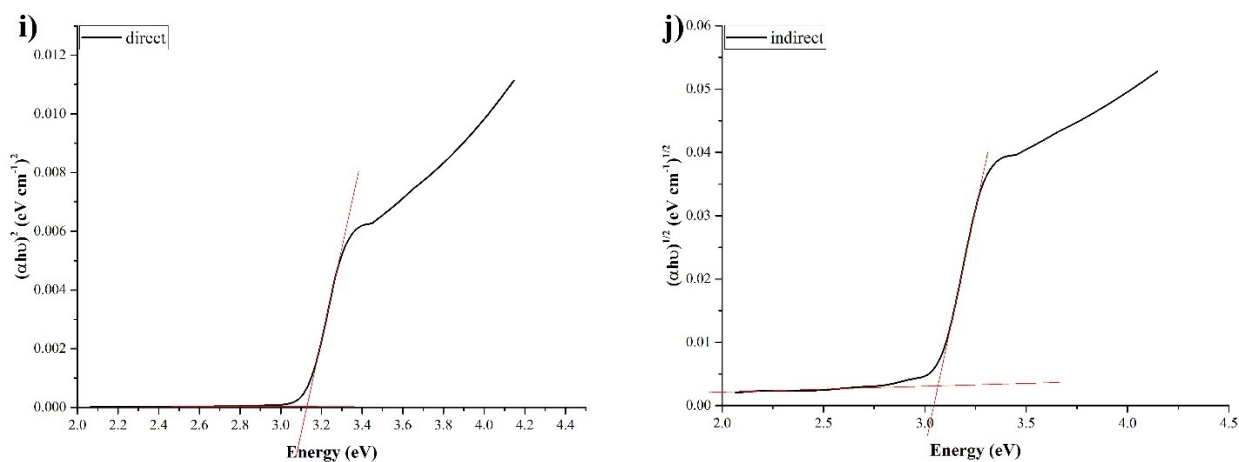


Figure S3. Direct and indirect band gap calculation graphs of (a-b) pure WO_3 , (c-d) 5% ZnO/WO_3 , (e-f) 10% ZnO/WO_3 , (g-h) 25% ZnO/WO_3 and (i-j) pure ZnO .

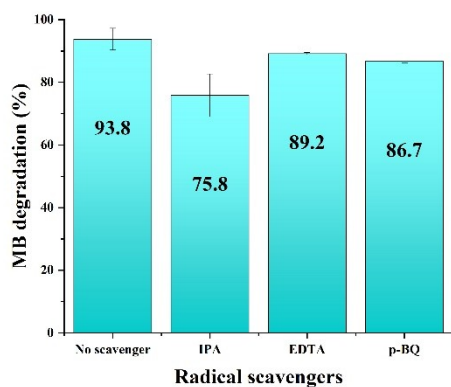


Figure S4. Radical quenching experiments (scavenger concentrations – 5 mM each).

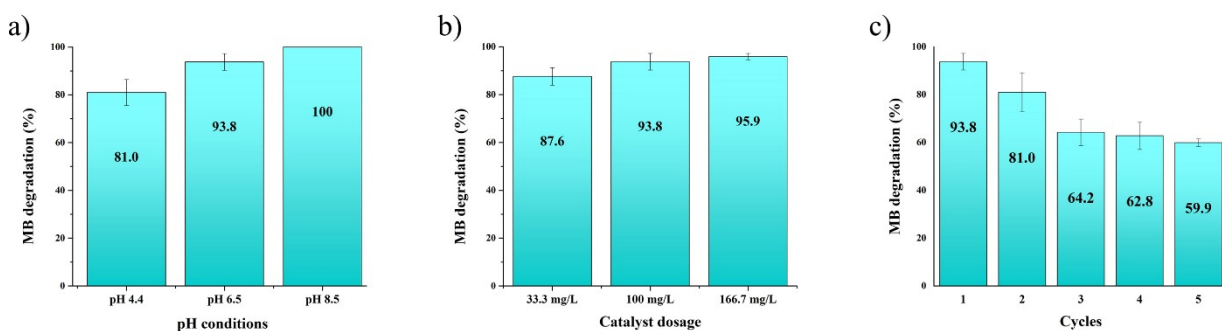


Figure S5. The effects of pH (a) and catalyst dosage (b) on MB degradation. Recyclability (c) of 5% ZnO/WO_3 catalyst.

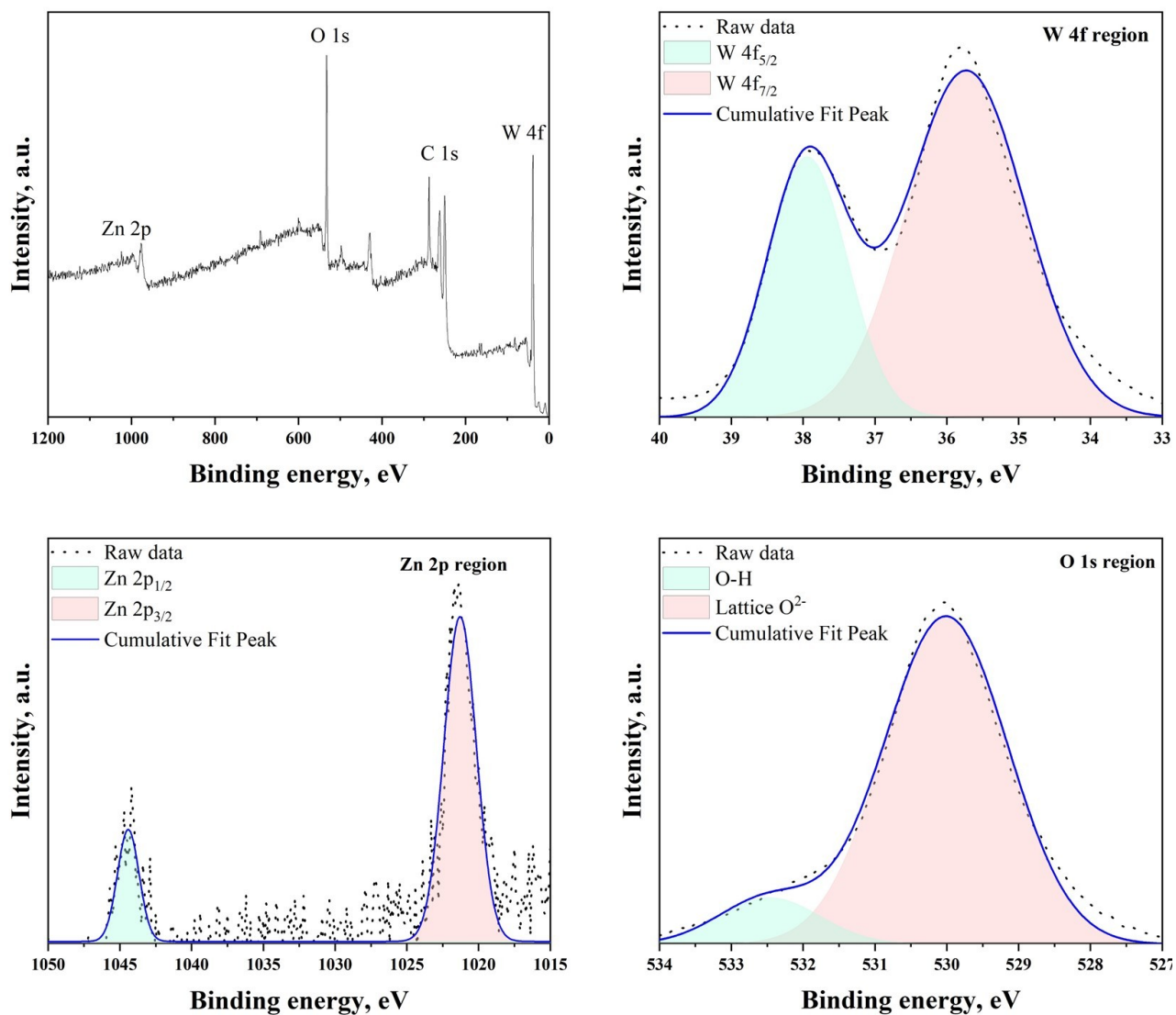


Figure S6. XPS survey and high-resolution spectra of the 5% ZnO/WO₃ composite after 5 cycles.